Valence Alignment in Human-AI Story Co-Creation

Analysis of Field Data and AI-AI Baseline Comparison

PENPAL Project

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0.1 Summary

- Emotional alignment is asymmetric: the model reliably aligns to the user (User → AI), while humans do not mirror the preceding AI tone.
- Field data shows greater variance in valence and weaker overall coordination compared to the simulated baseline.
- Overall, affective dynamics in the field are primarily model-driven rather than mutually convergent.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 870 17

## Simulated data dimensions: 30 12

##
## Field data columns:

##  [1] "timestamp"                 "user"
##  [3] "ai"                        "conversation_id"
##  [5] "respondent_id"             "interaction_count"
##  [7] "llm_type"                  "edit_distance"
##  [9] "turn"                      "starter"
## [11] "user_embedding"            "ai_embedding"
## [13] "pct_turn"                  "user_sentiment_score"
## [15] "ai_sentiment_score"        "user_sentiment_projection"
## [17] "ai_sentiment_projection"

##
## Simulated data columns:

##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===

## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              87     696                    8           -0.0891         0.00381

##
## === Simulated Data Summary ===

## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
```

```
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===

## Field data - Turn 1 with non-NA lag (should be 0): 0

## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

2 2. Descriptive Statistics

2.1 2.1 Token Count Statistics

```
## === Field Data Token Statistics ===

## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             25.4           5.28           29.1         7.45            0.913

##
## === Simulated Data Token Statistics ===

## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             26.0           2.17           41.1         3.73            0.636
```

2.2 2.2 Valence Distribution Statistics

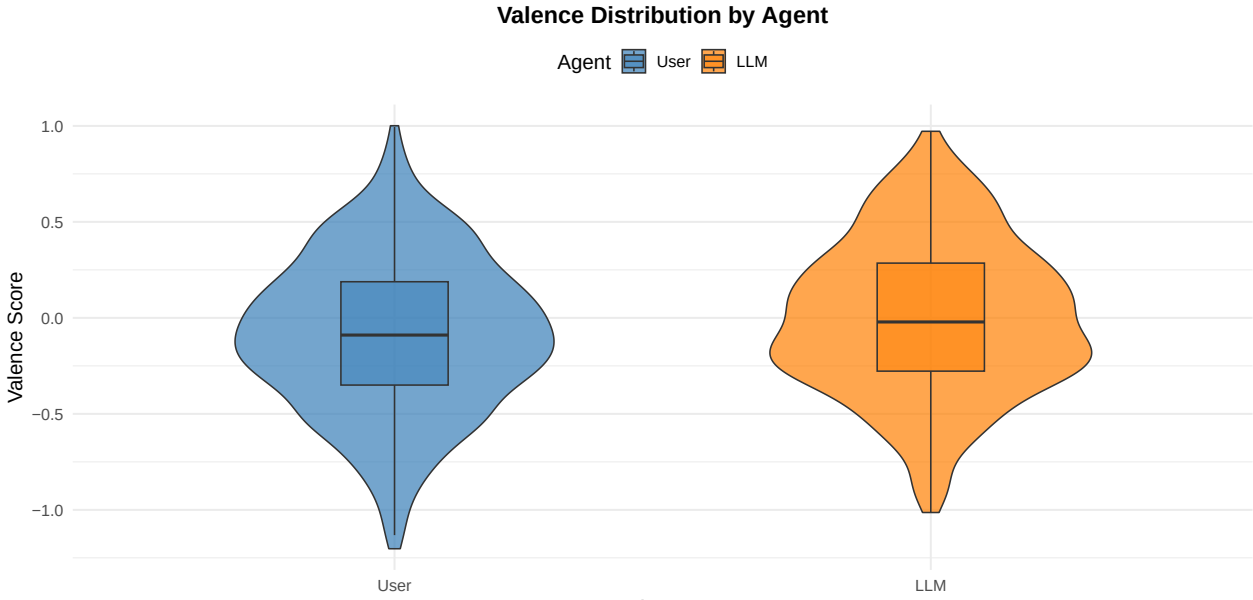
```
## # A tibble: 4 x 5
##   data_source agent mean_valence sd_valence median_valence
##   <chr>       <chr>        <dbl>      <dbl>          <dbl>
## 1 Field       ai         0.00381      0.397        -0.0212
## 2 Field       user      -0.0891       0.403        -0.0895
## 3 Simulated   ai        -0.296        0.681        -0.487
## 4 Simulated   user      -0.407        0.529        -0.504
```

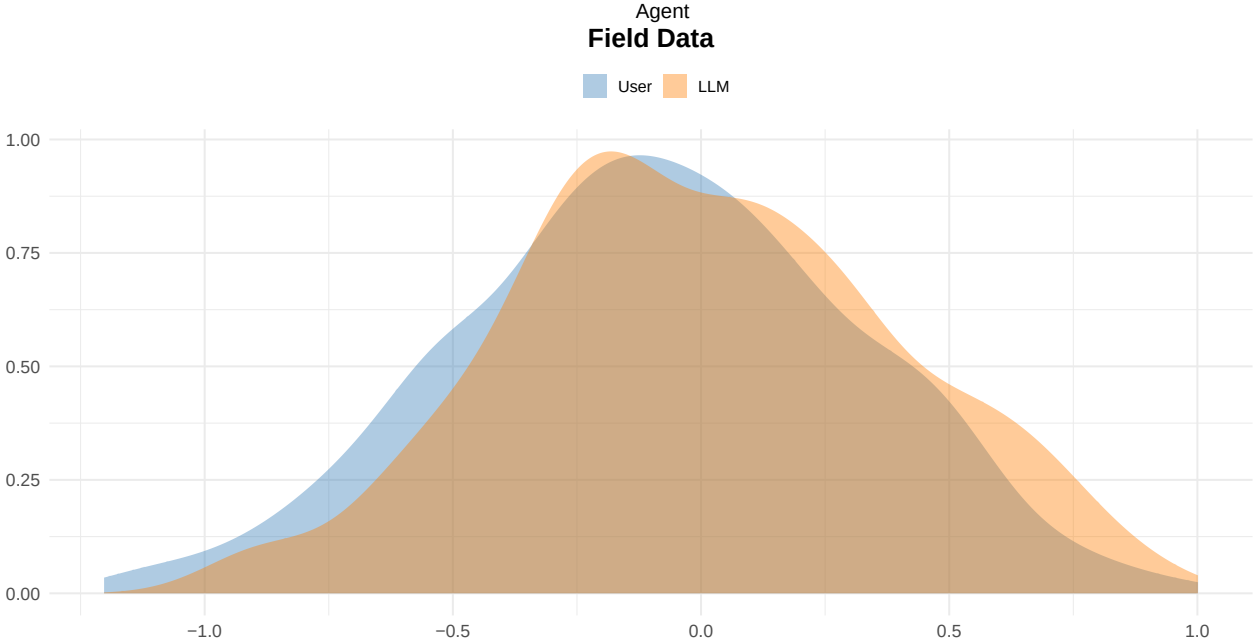
3 3. Valence Distribution Analysis

3.1 3.1 Valence Distribution Difference Test

```
## === Valence Difference Model Coefficients ===

##             Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.0891     0.0259 86.0015 -3.4428    9e-04
## agentLLM      0.0929     0.0214 86.0007  4.3447    0e+00
```

3.2 3.2 Combined Valence Distribution Plot





4 4. Valence Alignment Analysis

4.1 4.1 Per-Story Correlation (Same Turn)

```
## === Same-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.232  0.389      0.290        87
## 2 Simulated     -0.246  0.440     -0.314         6
```

4.2 4.2 Per-Story Correlation (Previous Turn)

```
## === Previous-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field         0.0908  0.395      0.123        87
## 2 Simulated     0.533   0.224      0.546         6
```

4.3 4.3.1 All turns story correlation (same turn)

```
## === Same-Turn All Turns Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.427     NA      0.427         1
## 2 Simulated     -0.110     NA     -0.110         1
```

4.4 4.3.2 All turns story correlation (Previous Turn)

```
## === Previous-Turn Aggregate Correlation ===
```

```
## Field: Turn 1 observations (excluded): 0
```

```
## Field: Valid lag observations: 609
```

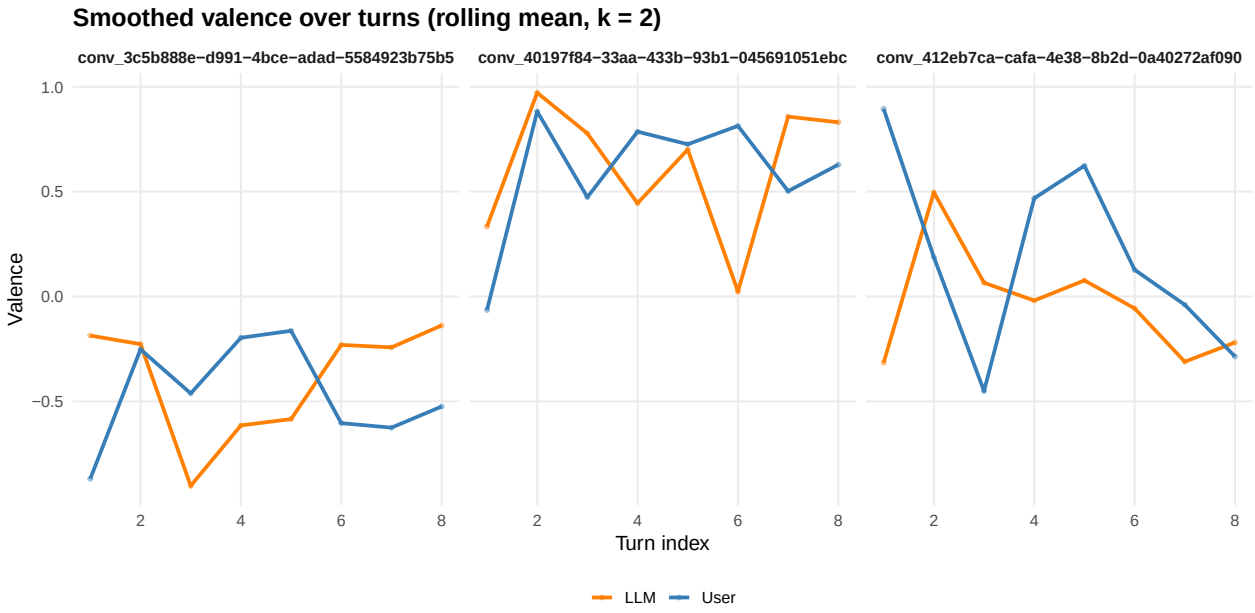
```
## Simulated: Turn 1 observations (excluded): 6
```

```
## Simulated: Valid lag observations: 24
```

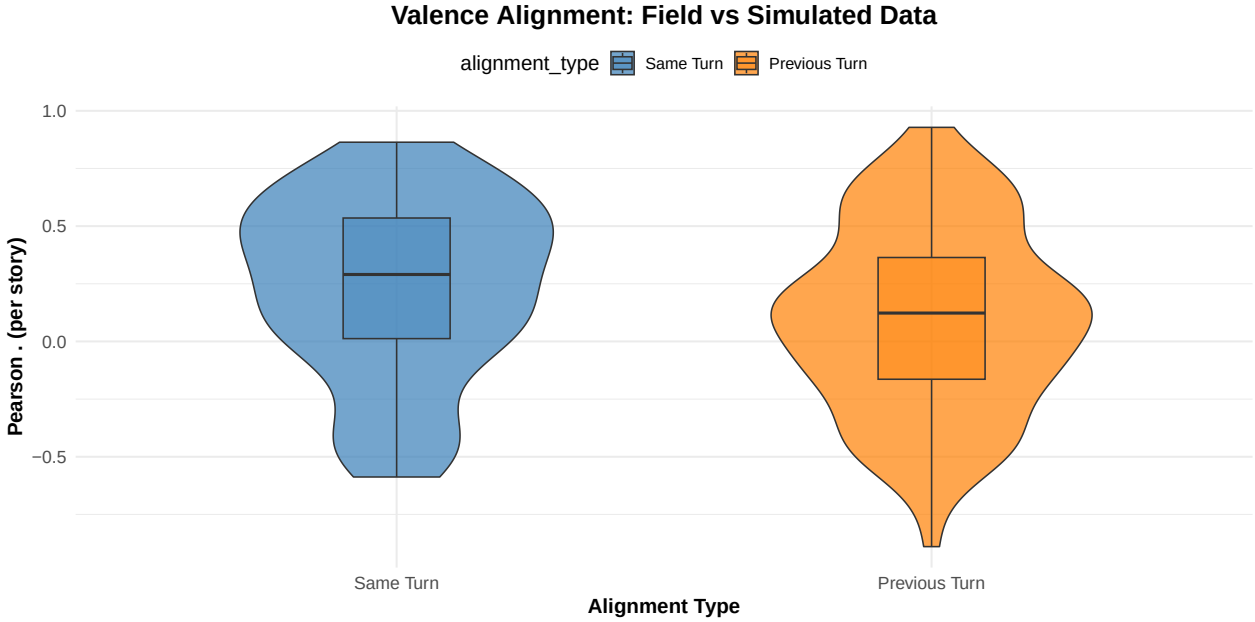
```
## === Previous-Turn All Turns Alignment Summary ===
```

```
## # A tibble: 2 x 6
##   data_source mean_rho sd_rho median_rho n_total n_valid
##   <chr>          <dbl>  <dbl>      <dbl>   <int>   <int>
## 1 Field          0.365     NA      0.365     696     609
## 2 Simulated      0.425     NA      0.425      30      24
```

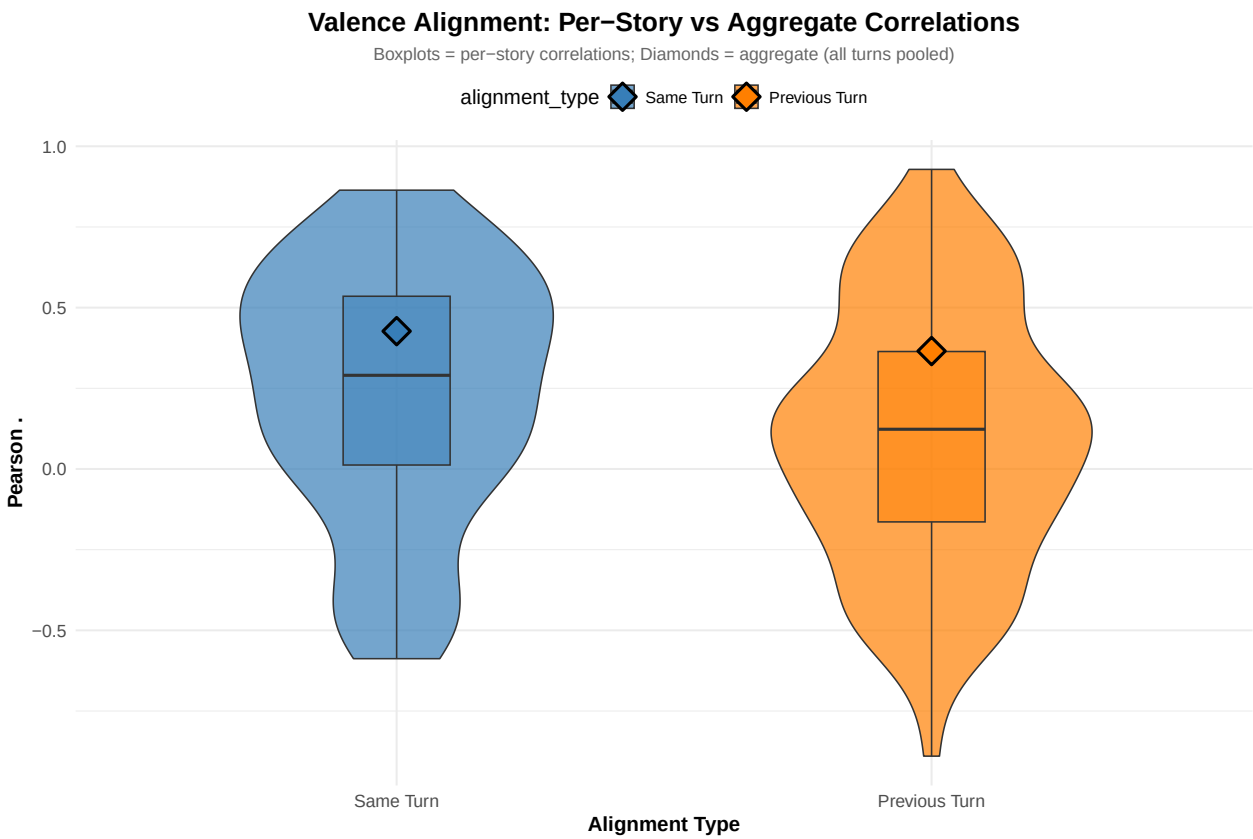
4.5 4.3.3 Probe Sentiment Narrative Development



4.6 4.4.1 Alignment Comparison Plot

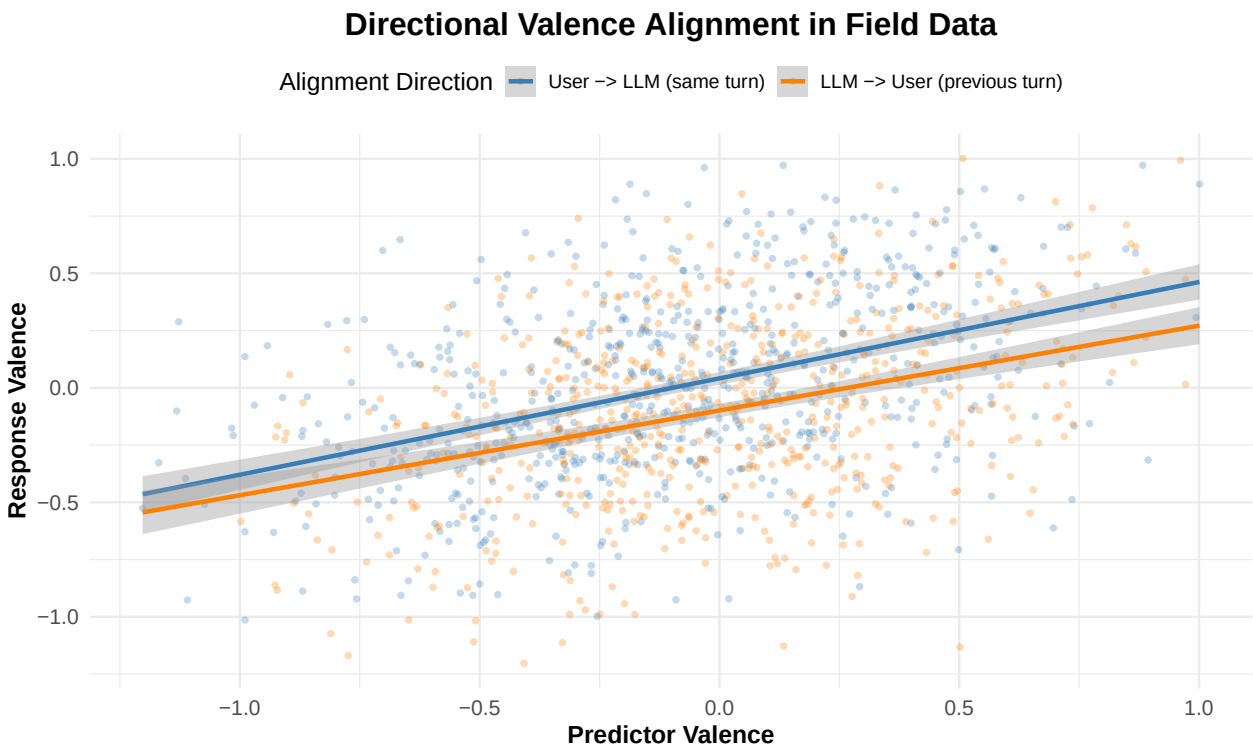


4.7 4.4.2 Comparison: Per-Story vs Aggregate Correlations




```
##
## === Per-Story vs Aggregate Correlation Comparison ===

## # A tibble: 4 x 4
##   data_source alignment_type `Per-Story Mean` `Aggregate All Turns`
##   <chr>       <fct>                    <dbl>                 <dbl>
## 1 Field       Same Turn               0.232                  0.427
## 2 Field       Previous Turn           0.0908                 0.365
## 3 Simulated   Same Turn              -0.246                 NA
## 4 Simulated   Previous Turn           0.533                 NA

## === Directional Alignment Long-Format Data ===

## # A tibble: 2 x 5
##   alignment_direction n_rows mean_predictor mean_response n_conversations
##   <fct>               <int>          <dbl>         <dbl>           <int>
## 1 User_to_AI            696       -0.0891        0.00381              87
## 2 AI_to_User            609       -0.00336      -0.100                87

## === Mixed Effects Model (Standardized): Directional Alignment (User -> AI vs AI -> User) ===

##                                          Estimate Std. Error   df t value
## (Intercept)                                0.0000     0.0348 1301  0.0000
## predictor_z                                0.2316     0.0372 1301  6.2322
## alignment_directionAI_to_User              0.0000     0.0509 1301  0.0000
## predictor_z:alignment_directionAI_to_User -0.1408     0.0547 1301 -2.5740
##                                          Pr(>|t|)
## (Intercept)                                1.0000
```

```
## predictor_z                                  0.0000
## alignment_directionAI_to_User                1.0000
## predictor_z:alignment_directionAI_to_User    0.0102
```



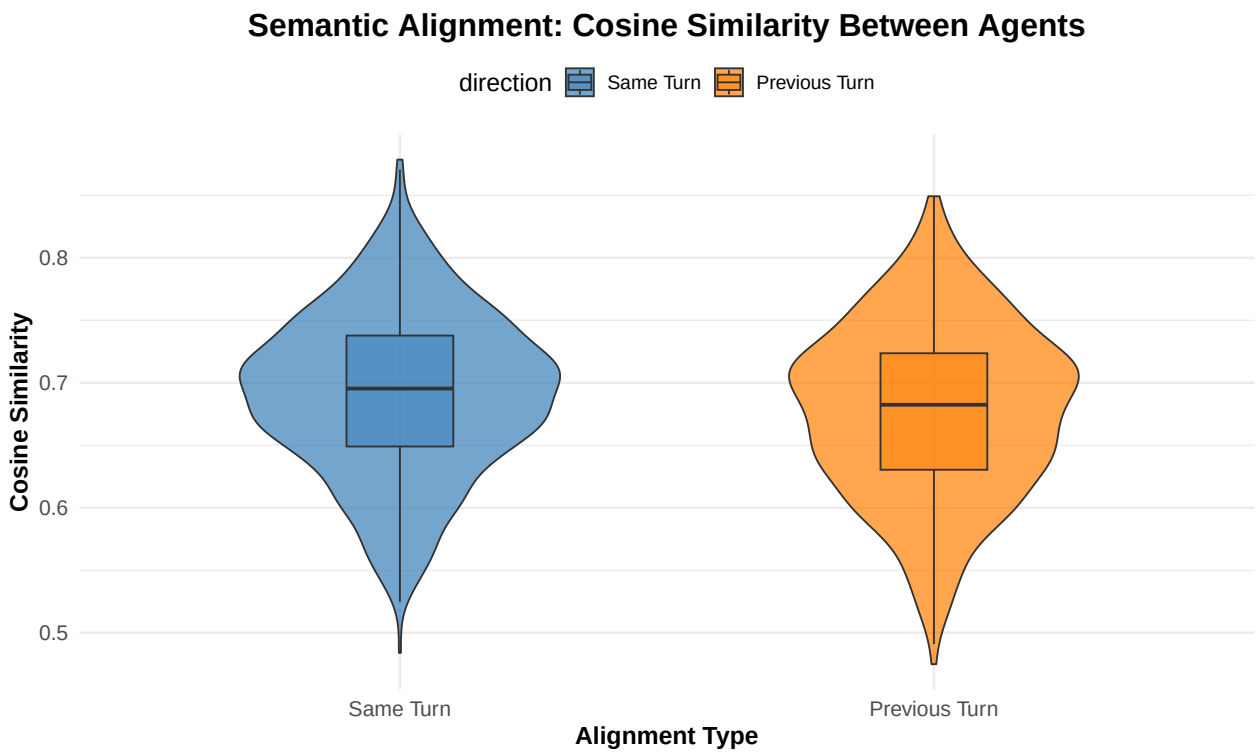
4.8 4.6 Statistical Tests: Alignment Differences

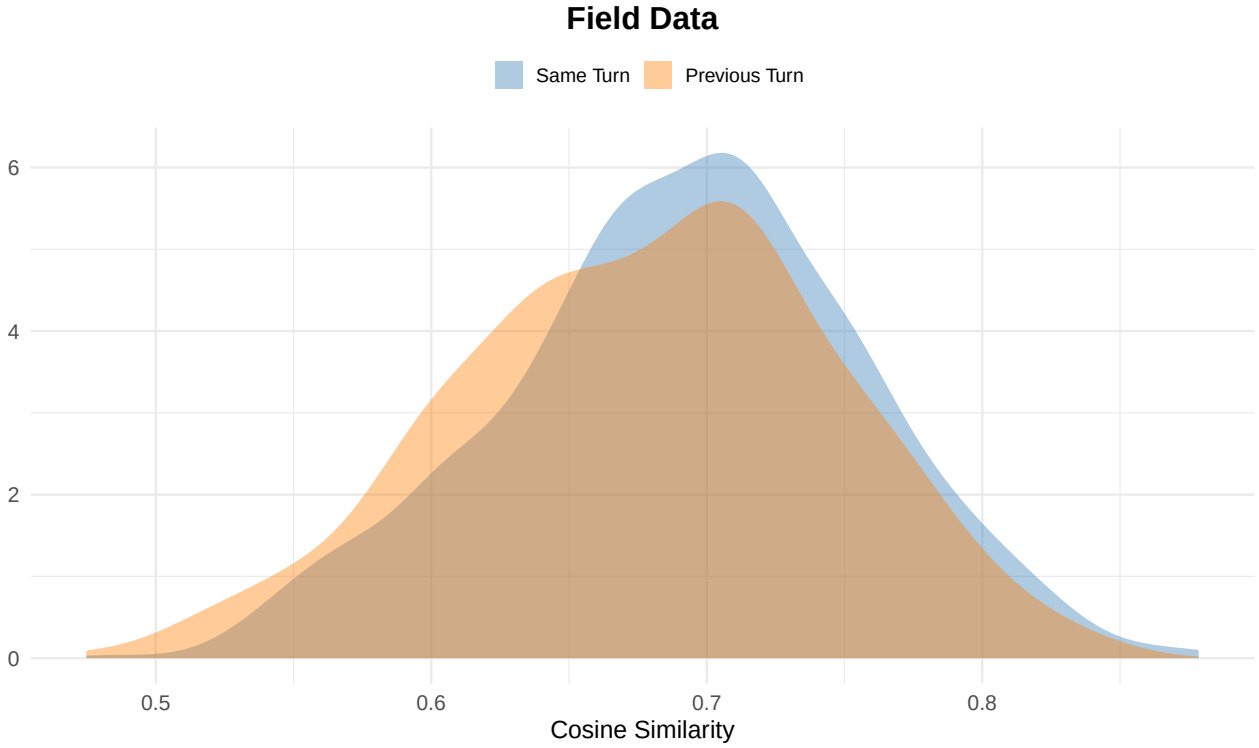
```
##
## === One-Sample Tests Against Zero ===

## # A tibble: 4 x 6
##   data_source alignment_type t_stat     p_value mean_rho     n
##   <chr>       <fct>           <dbl>       <dbl>    <dbl> <int>
## 1 Field       Same Turn        5.55 0.000000312   0.232     87
## 2 Field       Previous Turn    2.14 0.0349        0.0908    87
## 3 Simulated   Same Turn       -1.37 0.229        -0.246      6
## 4 Simulated   Previous Turn    5.84 0.00209       0.533      6

##
## === Paired T-Test: Directional Asymmetry (Same vs Prev) ===

## Field Data: Paired t-test (Fisher z-transformed Same Turn - Previous Turn)

##
##  Paired t-test
##
## data:  atanh(pairs_field$rho_same) and atanh(pairs_field$rho_prev)
## t = 2.2023, df = 86, p-value = 0.03032
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.01613391 0.31536192
## sample estimates:
## mean difference
```

```
##       0.1657479
##
## --- Interpretation (Back-transformed means) ---
## Mean Same-Turn r (User->AI): 0.272
## Mean Prev-Turn r (AI->User): 0.112
##
## Simulated Data: Paired t-test (Fisher z-transformed Same Turn - Previous Turn)
##
##  Paired t-test
##
## data:  atanh(pairs_sim$rho_same) and atanh(pairs_sim$rho_prev)
## t = -3.4821, df = 5, p-value = 0.01762
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -1.6343524 -0.2461251
## sample estimates:
## mean difference
##      -0.9402387
```

4.9 Cosine Distribution Difference Test

```
##
## === Cosine Sim: User (Self vs Interlocutor) ===
##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6897     0.0041 86.0000 169.6669   0.0000
## directionInterlocutor  -0.0117     0.0040 85.9997  -2.9192   0.0045
##
## === Cosine Sim: AI (Self vs Interlocutor) ===
##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6860     0.0041 86.0120 167.2791   0.0000
## directionInterlocutor   0.0054     0.0036 86.2026   1.5244   0.1311
##
## === Cosine Sim: General Direction Comparison ===
##                        Estimate Std. Error       df  t value Pr(>|t|)
## (Intercept)              0.6914     0.0036  94.1491 192.3986    0e+00
## directionPrevious Turn  -0.0134     0.0034 691.2310  -3.9250    1e-04
```

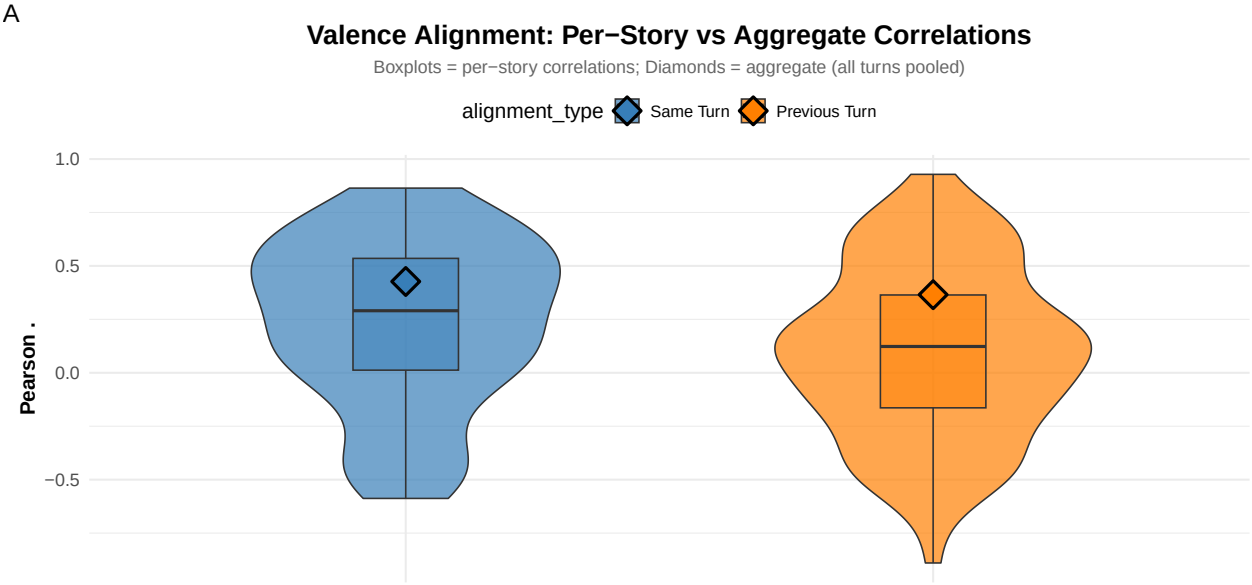




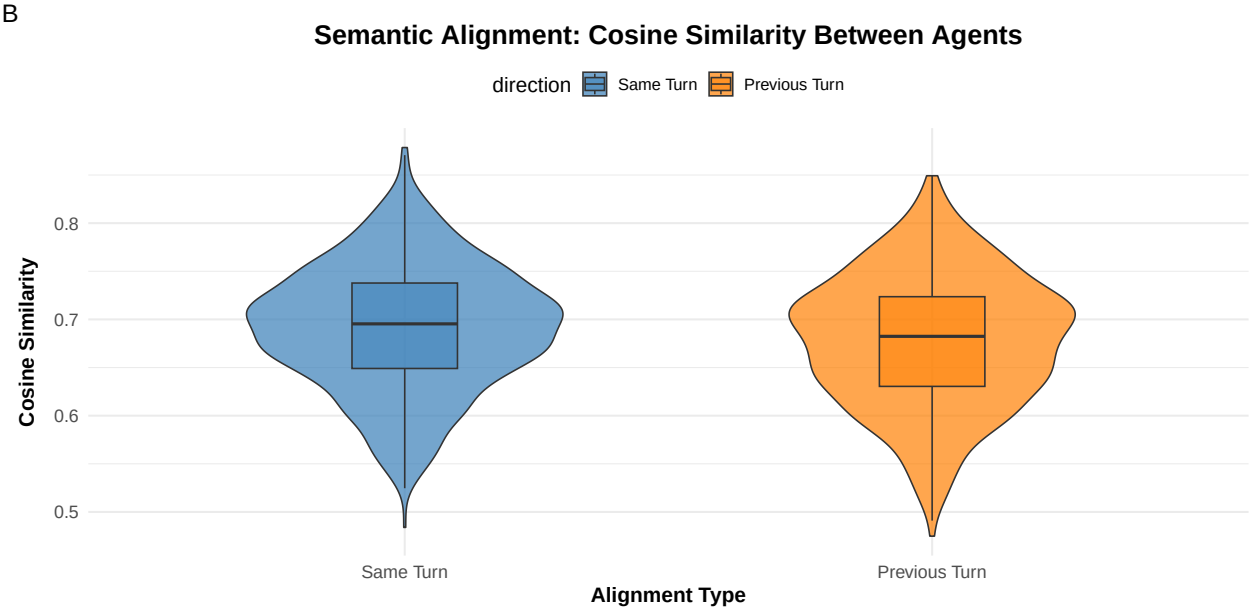
Combined Alignment Plot

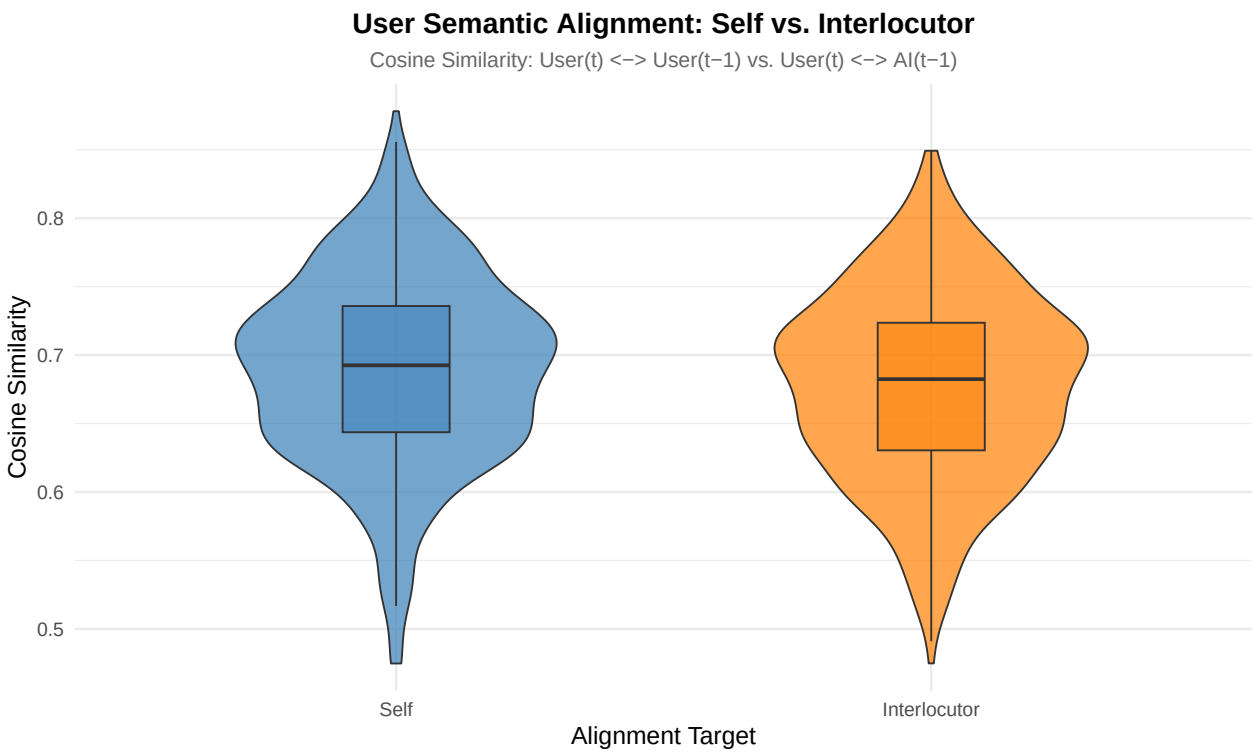
Comparison of Valence and Semantic Alignment

A



B





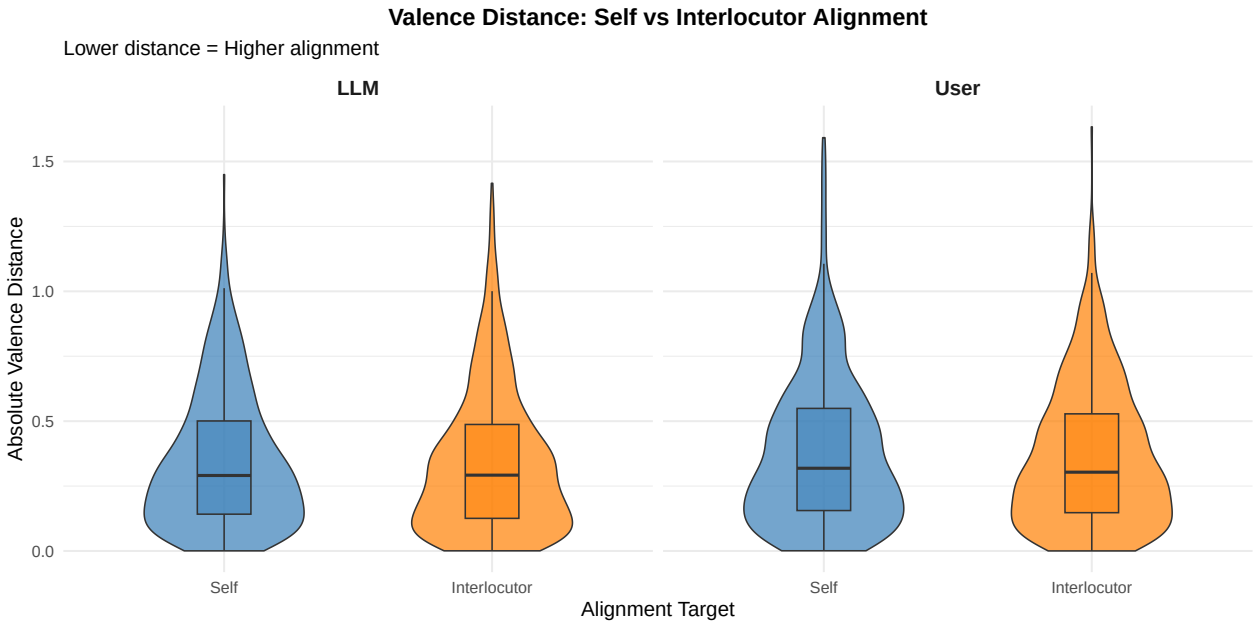
4.10 Valence Distance Difference Test

```
##
## === Valence Distance Model: User (Self vs Interlocutor) ===
```

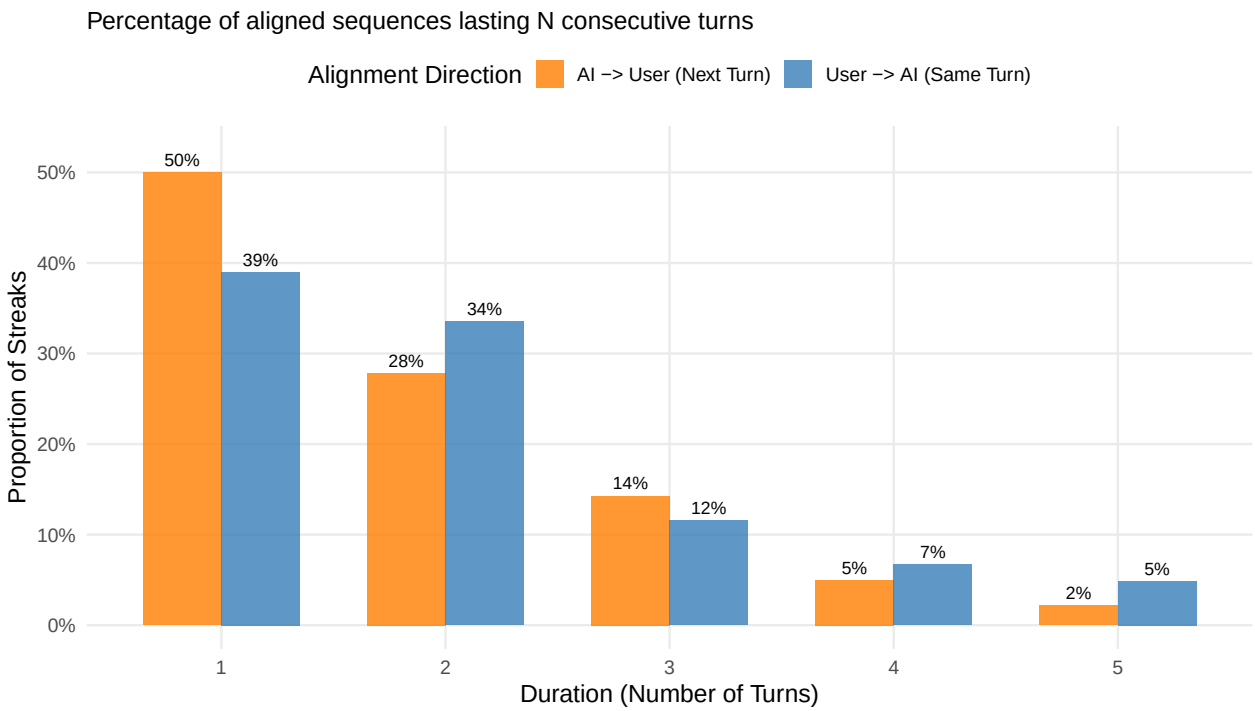
```
##                       Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)             0.3802     0.0145  94.256 26.2178   0.0000
## directionInterlocutor  -0.0126     0.0155 686.747 -0.8129   0.4166
```

```
##
## === Valence Distance Model: AI (Self vs Interlocutor) ===
```

```
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             0.3478     0.0106 1151.3136 32.7421   0.0000
## directionInterlocutor  -0.0056     0.0166   61.3057 -0.3381   0.7365
```



4.11 5. Duration of alignment




```
##
##  Welch Two Sample t-test
##
## data:  duration_list by direction
## t = -2.4879, df = 297.73, p-value = 0.0134
## alternative hypothesis: true difference in means between group AI -> User (Next Turn) and group User
## 95 percent confidence interval:
##  -0.63278418 -0.07383603
```

```
## sample estimates:
## mean in group AI -> User (Next Turn) mean in group User -> AI (Same Turn)
##                              1.835714                              2.189024
```

5 5. Summary and Conclusions

5.1 5.1 Key Findings Summary

```
## # A tibble: 5 x 3
##   Metric                      Field_Data Simulated_Data
##   <chr>                            <dbl>          <dbl>
## 1 N Stories                       87              6
## 2 Mean User Valence               -0.089         -0.407
## 3 Mean AI Valence                  0.004         -0.296
## 4 Same-Turn Alignment ( )          0.232         -0.246
## 5 Previous-Turn Alignment ( )      0.091          0.533
```

6 6. Supplementary: Interaction Effects

6.1 6.1 Starter Interaction Effects

6.1.1 6.1.1 Same-Turn Alignment by Starter (Removed for now)

```
##             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  -0.0389     0.0334  249.4249 -1.1633   0.2458
## turn         -0.0007     0.0040 1304.0000 -0.1724   0.8632
```

6.1.2 6.1.2 Previous-Turn Alignment by Starter (Removed for now)

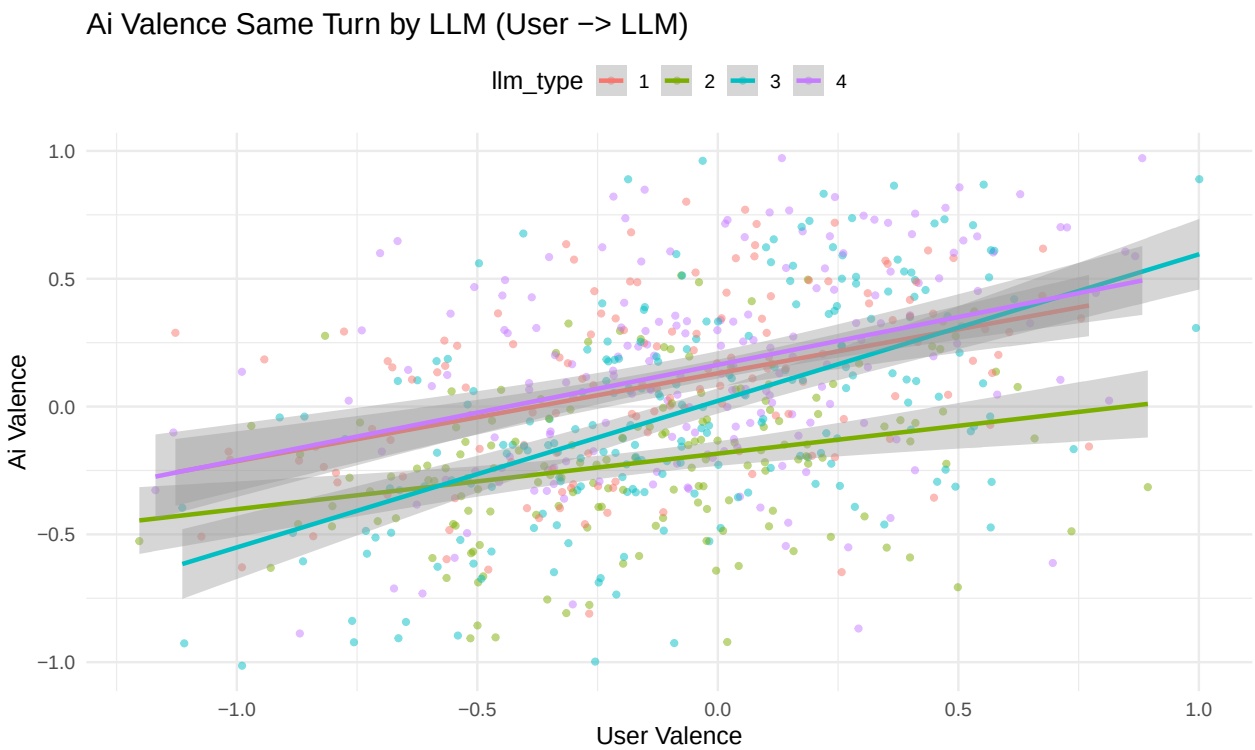
6.1.3 6.1.3 Per-Story Correlations by Starter (Removed for now)

6.2 6.2 LLM Type Interaction Effects

6.2.1 6.2.1 Same-Turn Alignment by LLM Type

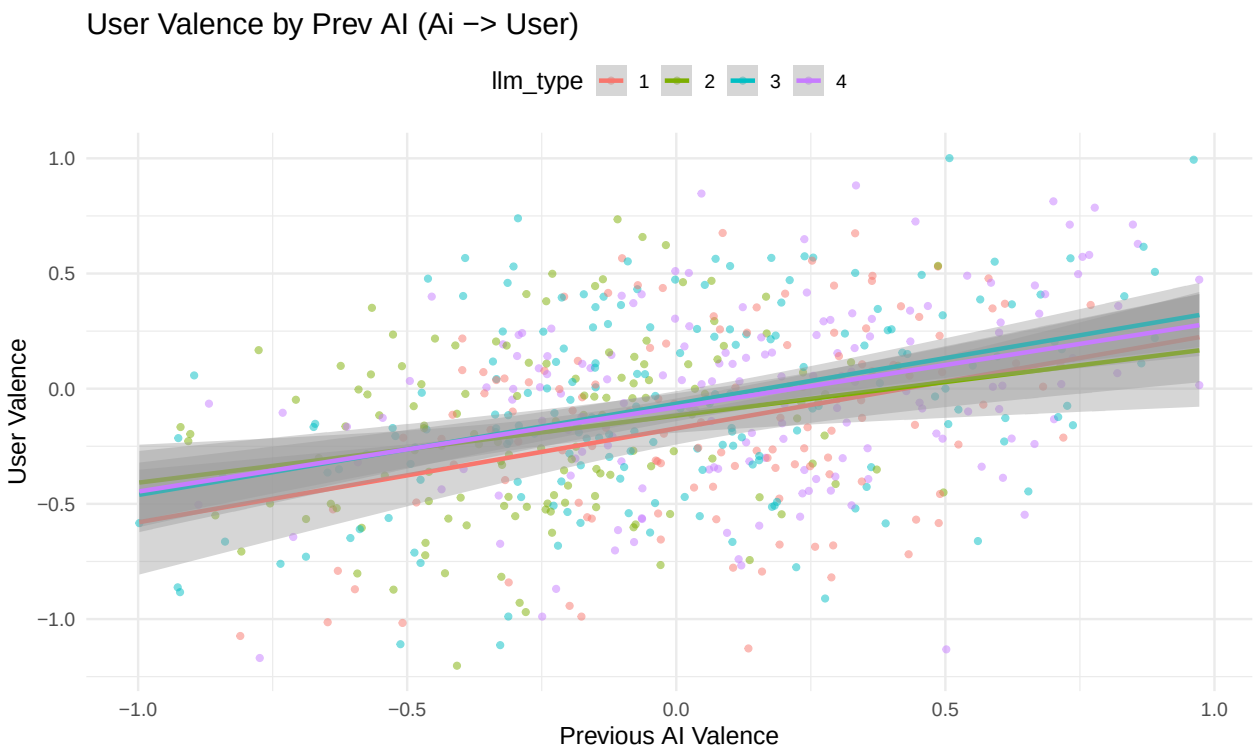
```
## === Same-Turn Alignment: LLM Type Interaction ===
```

```
##                                Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                      0.1208     0.0449  80.1881  2.6891   0.0087
## user_valence                     0.2722     0.0682 676.4800  3.9916   0.0001
## factor(llm_type)2               -0.3146     0.0623  82.2939 -5.0474   0.0000
## factor(llm_type)3               -0.1071     0.0581  78.6317 -1.8434   0.0690
## factor(llm_type)4                0.0399     0.0595  77.9823  0.6709   0.5042
## user_valence:factor(llm_type)2  -0.1164     0.0975 686.8242 -1.1946   0.2327
## user_valence:factor(llm_type)3   0.1684     0.0919 666.1010  1.8313   0.0675
## user_valence:factor(llm_type)4  -0.0032     0.0932 685.0224 -0.0347   0.9723
```



6.2.2 6.2.2 Previous-Turn Alignment by LLM Type

```
## === Previous-Turn Alignment: LLM Type Interaction ===

##                                   Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                        -0.1634     0.0515  74.5484 -3.1731   0.0022
## ai_valence_lag1                     0.2768     0.1031 584.9247  2.6850   0.0075
## factor(llm_type)2                   0.0243     0.0743  88.1319  0.3267   0.7447
## factor(llm_type)3                   0.0962     0.0667  73.6060  1.4412   0.1538
## factor(llm_type)4                   0.0965     0.0694  76.7147  1.3911   0.1682
## ai_valence_lag1:factor(llm_type)2  -0.0849     0.1457 600.0333 -0.5826   0.5604
## ai_valence_lag1:factor(llm_type)3  -0.0596     0.1246 581.6830 -0.4787   0.6323
## ai_valence_lag1:factor(llm_type)4  -0.0047     0.1298 583.9839 -0.0362   0.9711
```



6.2.3 6.2.3 Per-Story Correlations by LLM Type

```
## === Same-Turn Correlations by LLM Type ===
```

```
## # A tibble: 4 x 4
##   llm_type mean_rho sd_rho n_stories
##      <int>    <dbl>  <dbl>     <int>
## 1        1    0.301  0.353        18
## 2        2    0.198  0.364        20
## 3        3    0.255  0.405        26
## 4        4    0.180  0.432        23
```

```
##
## ANOVA (LLM type effect on same-turn rho):
```

```
##                  Df Sum Sq Mean Sq F value Pr(>F)
## factor(llm_type)  3  0.185 0.06177   0.399  0.754
## Residuals        83 12.846 0.15477
```

6.3 6.3 Valence Distribution by Conditions

6.3.1 6.3.1 Valence by Starter

6.3.2 6.3.2 Valence by LLM Type

```
## # A tibble: 8 x 5
##   llm_type agent        mean_valence sd_valence n_turns
##      <int> <chr>               <dbl>      <dbl>   <int>
## 1        1 ai_valence         0.0854      0.344     144
## 2        1 user_valence      -0.130       0.428     144
## 3        2 ai_valence        -0.219       0.296     160
## 4        2 user_valence      -0.159       0.381     160
## 5        3 ai_valence        -0.0146      0.425     208
```

```
## 6        3 user_valence      -0.0641      0.404     208
## 7        4 ai_valence         0.154       0.394     184
## 8        4 user_valence      -0.0246      0.393     184
```

6.4 7. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Pacific/Auckland
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] here_1.0.2      afex_1.4-1      cowplot_1.1.3   patchwork_1.3.0
##  [5] zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1  lmerTest_3.1-3
##  [9] lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3 forcats_1.0.0
## [13] stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2     readr_2.1.5
## [17] tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           plyr_1.8.9          abind_1.4-8
## [22] multcomp_1.4-26     withr_3.0.2         numDeriv_2016.8-1.1
## [25] grid_4.4.2          fansi_1.0.6         xtable_1.8-4
## [28] colorspace_2.1-1    emmeans_2.0.1       scales_1.3.0
## [31] MASS_7.3-61         tinytex_0.52        cli_3.6.3
## [34] mvtnorm_1.3-3       rmarkdown_2.28      crayon_1.5.3
## [37] ragg_1.3.2          generics_0.1.3      rstudioapi_0.16.0
## [40] reshape2_1.4.4      tzdb_0.4.0          minqa_1.2.8
## [43] splines_4.4.2       assertthat_0.2.1    parallel_4.4.2
## [46] vctrs_0.6.5         boot_1.3-31         sandwich_3.1-1
## [49] carData_3.0-5       car_3.1-3           hms_1.1.3
## [52] bit64_4.0.5         Formula_1.2-5       systemfonts_1.1.0
## [55] glue_1.7.0          nloptr_2.1.1        codetools_0.2-20
## [58] gtable_0.3.5        munsell_0.5.1       pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.5.1            textshaping_0.4.0
## [64] rprojroot_2.1.1     vroom_1.6.5         evaluate_1.0.3
## [67] lattice_0.22-6      Rcpp_1.0.13         coda_0.19-4.1
```

```
## [70] nlme_3.1-166        mgcv_1.9-1          xfun_0.54
## [73] pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.